Convergence to a Geometric Attractor in a Viable System Kernel

Executive Summary — June 2026

Luis Pineda R.
Independent researcher — Valparaíso, Chile
luis@tallerpineda.cl

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What We Built

We constructed a computational kernel (`vsf-nucleus`) based on Beer's Viable System Model (VSM) and tested whether a system governed by a minimal set of cybernetic axioms converges to a stable attractor in state space. The kernel's state is a point in $\mathbb{R}^{128}$, organised into semantic blocks: identity, resources, entropy, operations, communication channels, attractor memory, and regulatory intelligence.

We ran over thirty-five experiments on physical hardware (Intel Core i7-7600U, Proxmox hypervisor) from April 2026, measuring convergence, basin geometry, thermal resilience, observer effects, architectural ablation, and — in the final experiment — the complete system assembled with all VSM layers active.

In June 2026 we coupled the kernel to a first genuinely external environment: a distributed climate network (`EXP-S1-CLIMA-01`) of twelve zones across Chile and the southern Pacific islands, operating as real S1 units on live ERA5 data. On that corpus we trained Markov chains per zone (30 years of historical data, 1991–2020), producing probabilistic excursion projections at $t{+}7$ and $t{+}30$ days, published via NATS every hour.

Main Results

1. The attractor is a stable point sink, confirmed by three independent methods. We measured the largest Lyapunov exponent directly (Benettin's method): $\lambda_{\max} = -0.010050 = \ln(0.99)$ per tick, at machine precision — negative, so nearby trajectories contract rather than diverge. A Monte-Carlo estimate places the basin volume at ≈ 1.0 over the reachable region, and persistent homology of the topology gives $\beta_0 = 1,\ \beta_1 = 0$ (one component, no hole, no limit cycle). Dynamics, volume, and shape agree: a single stable point.

2. A coordinate-free viability certificate, $V = \min(g, c, t, m)$. We distilled the measurements into a functional built from intrinsic invariants (feasibility, contraction sign, topological signature, noise margin). In this kernel the binding axis is $g = 0.102$ (the CPU budget constraint), and breaking any single axis drives $V \to 0$ independently — making V a discriminating vector rather than a cosmetic scalar always near one.

3. The observer is constitutive, not passive. Structured observation measurably reduces system entropy by 0.31 bits and compresses effective dimensionality from ≈ 5.0 to 3.9 — a

measured instance of second-order cybernetics, in which the observer is part of the dynamics that stabilises.

4. Anticipation requires external variance. Because the attractor keeps its viability measure invariant even under sustained CPU overload, S4 cannot read an early warning of its own regulated state: the signal is flattened by the same homeostasis that protects the kernel. Coupling with `EXP-S1-CLIMA-01` provides the first genuinely external variance. The Markov projections deliver to S4 a $P(\text{excursion})$ per zone that the internal attractor cannot generate on its own.

Key Finding: Two-Layer Viability

The ablation experiments reveal a **two-layer viability structure**:

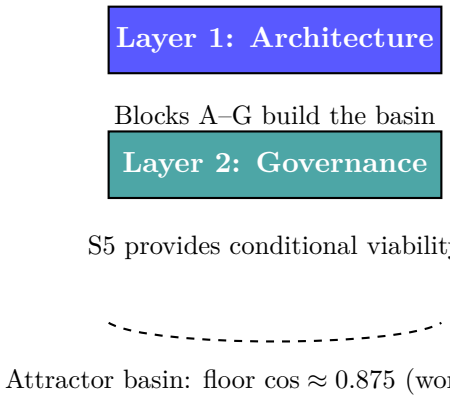


Figure 1: Architecture builds the basin and its homeostasis; governance is the difference between surviving stress and collapsing under it.

Layer 1: Architecture The arrangement of VSM blocks physically determines which regions of state space are habitable. The bare substrate (without blocks) drifts off-target and never stabilises; the blocks are what creates the point sink in $\mathbb{R}^{128}$.

Layer 2: Governance (S5) At rest, governance is *redundant* — the system converges with or without it. Under stress it is decisive. Governance has two arms: a *reactive* one (the algedonic safety response that regulates runaway thermal load) and a *deliberative* one (the constitutional veto of the covenant, which can halt the system's own autonomy). The experiments exercise both.

With both layers present the system sustains viability at cos ≈ 0.875 (on a 0–1 scale, with 0.70 as the viability threshold) even under maximum impulsive stress — a margin of 0.175 never breached in any experiment.

The Closure: an Operationally Closed System

The final experiment ran the complete kernel with all five Beer systems (S1–S5) active under a single perturbation. Each level acted within its own cybernetic scope. The decisive observation traverses the full model in a single tick: operations (S1) registered a drop, coordination (S2) delivered a policy signal, control (S3) detected the degradation and *attempted* a correction, and policy (S5) *vetoed* it — after which the system self-recovered. A separate test confirmed the constitutional veto end-to-end: it halts autonomous control under sustained degradation and restores it only on an explicit, cryptographically recorded human resumption.

This is the sense of the architecture: not a simulation of Beer's diagram, but a system in which every arrow carries a measured event.

Empirical Results

Property	Measurement	Status
Lyapunov exponent	$\lambda_{\max} = \ln(0.99) < 0$	Measured, sink
Basin volume	≈ 1.0 (reachable region)	Monte-Carlo
Topology	$\beta_0 = 1,\ \beta_1 = 0$	No limit cycle
Basin uniqueness	$\cos_{\text{avg. paired}} = 0.9854$	Confirmed
Thermal viability	viable at 68 °C	Sustained
Basin floor	$\cos \approx 0.875$	Margin 0.175
Entropy drop	$\Delta H = 0.31$ bits	Constitutive
Viability certificate	$V = \min(g, c, t, m),\ g = 0.102$	Discriminates
Process governance	4 OS processes, $\tau = 1.0$	cgroup v2 live
VSM operational closure	S1–S5 causal in one run	Demonstrated
External coupling (S1)	12 zones, ERA5 live data	Operational
Markov projections (S4)	$P(\text{exc.}_{t+7d})$ per zone, NATS	Operational

What This Means

1. **Viability is geometric.** The system's stability emerges from the geometry of its state space, not from an external imposition. This is what Beer meant by a *viable system*.
2. **Ethical constraints enter as feasibility conditions.** The axioms prohibiting harm and coercion are not added on top — a covenant violation collapses viability ($V \to 0$) exactly as exhausting the CPU budget does. The mathematics is *indifferent* to what the constraints encode ethically; that indifference is precisely what makes it a general certifier rather than a moral theory.
3. **Governance is conditional resilience.** A good policy layer does not *create* viability — architecture does — but it is the difference between graceful survival and collapse when the system is challenged.
4. **The basin is deep.** Under every perturbation tested the system never approached the viability threshold (0.70); the closest approach was 0.875, a margin of 12.5% above failure.
5. **Anticipation requires environment.** A fully homeostatic system cannot anticipate itself: the regularity that stabilises it is the same regularity that flattens the anticipatory signal. External variance — in this case, real climatic variability across twelve zones — is a structural requirement of the S4 channel, not a design option.

Honest Limits

The work runs on a single hardware node; the thermal stressor, although it raises the actual silicon temperature, is a synthetic load; and the convergence experiments are closed-system runs. The external coupling is operational, but the attractor's response curve to sustained long-term signals is still work in progress: we have the projections, but not yet a measure of how the kernel integrates them under real climatic variability maintained over weeks or months.

The distributed architecture (eight LXC containers on a single Proxmox hypervisor) is internal to the same system it measures: the resilience results hold under that hardware profile, and extension to multiple nodes or heterogeneous workloads (databases, inference) is future work.

Next Steps

1. **Attractor response to sustained external environment.** With the climate network operational and the Markov projections online, the open question is how the attractor geometry responds to signals maintained over weeks: does the centroid shift? does the basin volume change? do excursions appear that the closed attractor does not produce?
2. **Inter-system coordination via a formal substrate.** A parallel line of work studies whether two heterogeneous models can coordinate as distinct VSM layers through a shared formal language (VSL) — the basis of a second paper.
3. **vOS on bare metal.** Deploying the kernel against real workloads (databases, inference) to measure attractor geometry under diverse load profiles and estimate the Landauer boundary of computational viability.

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For the Beer Community

This work returns to first principles: *what does it mean for a system to be viable?* We give a computational answer grounded in cybernetic axioms, using instruments — Benettin's algorithm, persistent homology, cryptographic attestation, Markov chains over climatic history — that did not exist when Beer wrote. The architecture–governance distinction, and the demonstration that all five systems close an operational loop on real hardware, suggest that the VSM blocks are not metaphors: they are viability requirements with measurable geometry.

There are questions this work cannot answer on its own. Does the architecture–governance distinction have a counterpart in real organisational systems, or is it an artefact of the computational implementation? Is the threshold $\cos \geq 0.70$ an invariant that makes sense for heterogeneous systems (organisations, networks, institutions), or is it specific to this kernel? What does someone who has worked with VSM in the field see when they read the ablation curve?

The full paper (≈70 pages), data, and experimental logs are available on request. So is the conversation.